

Question	Answer	Marks
5(a)	number of particles per unit amount of substance	B1
5(b)	<u>any three points from:</u> - the gas is ideal - volume (of gas) = 0.46 m^3 <u>up to two points from:</u> <i>either:</i> mass of one molecule (of gas) = $6.6 \times 10^{-27} \text{ kg}$ <i>or:</i> mass of one mole (of gas) = $4.0 \times 10^{-3} \text{ kg}$ <i>either:</i> amount of gas = 2.6 mol <i>or:</i> number of molecules (of gas) = 1.6×10^{24} <i>either:</i> mass of the gas = 0.010 kg <i>or:</i> density of the gas = 0.022 kg m^{-3}	B3
5(c)(i)	<i>either:</i> gas is ideal so yes <i>or:</i> gas obeys $pV \propto T$ so yes <i>or:</i> gas obeys $\langle c^2 \rangle \propto T$ so yes <i>or:</i> gas obeys $E_K \propto T$ so yes	B1

Question	Answer	Marks
5(c)(ii)	<u>any two points from:</u> • molecules are in <u>constant</u> random motion • (all) collisions (involving molecules) are elastic • no intermolecular forces (except during collisions) • (all) collisions (involving molecules) are instantaneous	B2